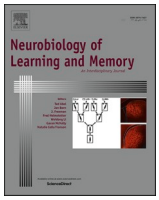
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Hormonal regulation of behavioral and emotional persistence: Novel insights from a systems-level approach to neuroendocrinology☆

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ABSTRACT

Gonadal sex steroid hormones regulate internal states, social drive, perception of external cues, and learning and memory. Fluctuating hormones influence mood and emotional states, enabling flexibility in instinctive behaviors and cognitive decisions. Conversely, elevated hormone levels help sustain emotional states and behavioral choices, ensuring the precise execution of costly social behaviors within optimal time windows to maximize reproductive success. While decades of work have shed light on the cellular and molecular mechanisms by which sex hormones alter neural excitability and circuit architecture, recent work has begun to tie many of these changes to principles of computation using the tools of systems neuroscience. Here, we will outline the mechanisms by which sex steroid hormones alter neural functioning at the molecular and cellular level and highlight recent work that points towards changes in specific computational functions, including the generation and maintenance of neural and behavioral persistence.

1. Introduction

While hormones have long been associated with powerful changes in mood, internal state, and behavior, recent work has begun to deepen our understanding of how these changes occur at both the mechanistic and computational levels. Sex hormones, which include estrogens, progesterone, and androgens, are present in varying amounts in both males and females, and circulating levels can change as a result of development, endogenous cycling, or behavior itself (McEwen and Milner, 2017; Gurvich et al., 2018; Jennings and de Lecea, 2020). Recent work focused on understanding the computational properties of sex hormones has frequently focused on females. This is apt, considering that hormonal transitions throughout a woman's lifespan are closely linked to changes in emotional states as seen in mood and anxiety changes during premenstrual, postpartum, and perimenopausal stages (Barth et al., 2023; Kundakovic and Rocks, 2022). Women face twice the risk of developing persistent negative emotional states, including anxiety and depression compared to men, with cycling sex hormones identified as a key biological factor contributing to this disparity (Cover et al., 2014; Eliot et al., 2023). Hormone withdrawal has also been shown to induce depressive symptoms and negative emotional states including reduced sociability in mice (Tao et al., 2023).

Exogenous alteration of hormonal profiles is also widespread. Millions of people globally use oral contraceptives (Lacasse et al., 2024), and menopausal women as well as gender-diverse individuals rely on hormone therapies (Aghi et al., 2022; Camon et al., 2024). Breast cancer survivors also rely on anti hormonal therapies, such as estrogen receptor degraders or aromatase inhibitors, to prevent recurrence (Ingraham et al., 2022). Many of these compounds cross the blood–brain barrier, impacting hormone-sensitive brain regions. Estrogens can also act as antidepressants or enhance the efficacy of existing treatments (Dossat et al., 2018). The effects on mood and emotional state are complex and influenced by several factors (Schuh et al., 2024) such as hormone formulation, dosage, schedule, and the individual's psychiatric history. Understanding the impact of these exogenous hormones on brain and emotional states is crucial for creating tailored formulations, empowering informed decision-making, improving gender-affirming care, and enhancing mental health for a significant portion of the population.

2. Higher estrogen states are associated with persistence and less behavioral variability

In animal models, sex steroid hormones have been primarily studied in the context of reproduction and reproductive behaviors, but

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mounting evidence suggests that behavior changes associated with hormones extend beyond this limited context. However, across reproductive and nonreproductive contexts, a common thread has emerged in both sexes that links higher levels of estrogens and androgens with behavioral persistence, and low levels of hormones with behavioral variability.

During reproductive-related behavior, female rodents display sexual receptivity or rejection based on hormonal state (Gutierrez-Castellanos et al., 2025; Inoue et al., 2019; Yin et al., 2022), with increased estrogen favoring receptivity during the *peri*-ovulatory stages of the estrous cycle such as proestrus and estrus, and more rejective behaviors observed during low estrogenic stages such as metestrus and diestrus. Proceptive behaviors, such as actively seeking and initiating interactions with male conspecifics, increase during high estrogenic states (Yin and Lin, 2023). Hormonal states also influence conspecific preferences and social choices, with more male preference in high estrogenic states (McHenry et al., 2017; Wang et al., 2025; Yin et al., 2022). This is consistent with the observation of more persistent and escalating sexual arousal during receptive behaviors (Liu et al., 2024). Female conspecific preferences shift from neutral or variable during diestrus to stronger and more stable male preferences during estrus (McHenry et al., 2017; Wang et al., 2025).

Fertile states are marked by heightened locomotor activity (Krause et al., 2021), reduced food intake (Ingraham et al., 2022), less sleep, and more awake time (Dorsey et al., 2020), potentially redirecting energy and attention towards mate-seeking. In these states, females also exhibit reduced anxiety-related behaviors (Kundakovic and Rocks, 2022) and dampened stress reactivity (Albert and Newhouse, 2019). In rats, high estrogen levels are further associated with enhanced sensitivity to risk (Orsini et al., 2021) and reward (Golden et al., 2023). During fertile stages, they display greater sensitivity to prior reward history, suggesting improved reward shaping, enhanced learning from past rewards, and less erratic decision-making (Golden et al., 2023). Similarly, in humans, menstrual cycle phases influence behavior. During the follicular phase, when estrogen levels are high, individuals show reduced response variability (Sumner et al., 2018), decreased bias for immediate rewards (Smith et al., 2014), and value higher average payoff over time (Lazzaro et al., 2016). In addition, emotional processing and reactivity to facial expressions vary with hormonal state, with individuals in the follicular phase exhibiting heightened socio-emotional sensitivity (Derntl et al., 2008; Macrae et al., 2002).

Sensory perception and environmental coding is also more stable when estrogen levels are high, which may help to guide individuals to their preferred social targets. Female mice in estrus display stable attraction towards male pheromones, but show indifference towards some of those cues in diestrus (Dey et al., 2015). Estrogen priming in ovariectomized female mice also induces attraction towards male odors (McHenry et al., 2017). During phases of elevated estrogen, female mice exhibit stable and enhanced auditory sensitivity and responsiveness to socially relevant sounds such as pup vocalizations (Banerjee and Liu, 2013; Caras, 2013). While estrogenic modulation of female's preference for male ultrasonic vocalizations remains to be fully understood in mice (Asaba et al., 2014), female songbirds have been shown to display enhanced selectivity for conspecific male vocalizations and more frequent song-evoked mating-ready displays with increased systemic estrogen during the breeding season (Caras, 2013). Song playbacks and social interactions, in turn, trigger rapid local estrogen synthesis in the songbird auditory forebrain, further enhancing auditory processing and auditory memory consolidation (Vahaba and Remage-Healey, 2018). Although the estrogenic modulation of social-touch perception is less clear, evidence in rats suggests that high estrogen levels increase somatic sensitivity relevant to lordosis posturing and reduce pain sensitivity (Reed et al., 2009). Furthermore, environmental representations are more stable during proestrus, supporting more robust spatial memory (Wolcott et al., 2025). Elevated estrogen levels in female mice have also been linked to fewer lapses in spatial working memory and improved object and social recognition (Taxier et al., 2020).

Beyond estrogen, hormonal change not associated with the estrous cycle can also influence behavioral persistence and offer evolutionary advantages. Maternal behavior in mice differs significantly from virgins: mothers exhibit sustained parental care, while virgins often display variable behavior, including infanticidal tendencies toward pups (Mei et al., 2023; Bailey and Isogai, 2022). Lactating mothers also show heightened maternal aggression toward intruders (Liu et al., 2022). In males, testosterone regulates dominance and aggression (Stagkourakis et al., 2020). Social interactions, such as encountering potential mates or challenges, can trigger acute testosterone surges (Jennings and de Lecea, 2020), and these testosterone pulses are believed to modify future behaviors, promoting persistent aggressive states (Guthman and Falkner, 2022) in order to defend territory or enhance competitive success. Testosterone levels in male mice are also thought to influence the production of ultrasonic vocalizations towards females (Asaba et al., 2014), with dominant males producing more frequent and persistent vocalizations when encountering a female (Wang et al., 2011). Future research is needed to map these changes in behavioral persistence to change in circuit and population level computation.

3. Hormones shape brainwide networks governing social behavior, cognition and perception

In many mammals, the estrous cycle in females is defined by the well-described activity of the hypothalamic-pituitary–gonadal (HPG) axis. Briefly, GnRH neurons in the hypothalamus neurons trigger the release of luteinizing hormone (LH) and follicle stimulating hormone (FSH) from the anterior pituitary, which in turn triggers a surge in gonadally produced estrogen followed by a surge in progesterone. The timing of the surge and then steep decline of estrogen is associated with ovulation and peak fertility. During these surges, estrogen and progesterone feedback to the hypothalamus to negatively regulate its own production. However, this hormonal feedback loop affects not only the activity related to hormonal production in the HPG axis itself, but also in populations of neurons brainwide that express receptors for these hormones with widely varying functions (Fig. 1).

Hormonal regulation of neurons can act through genomic and non-genomic pathways on neurons expressing hormone receptors. In the "classic" genomic pathway, estrogens and progesterone primarily act on neurons by binding to nuclear receptors (ERα/β, PRA/B), which dimerize and bind to specific genomic response elements, recruiting transcriptional machinery to regulate gene expression (Nilsson et al., 2001). Recent data has shown that genes under the control of estrogen have been shown to include those involved in neural wiring and synaptic maintenance (Gegenhuber et al., 2022). Additionally, these hormones can operate via "non-classical" pathways, triggering rapid changes in excitability through membrane-bound receptors that act via G-proteins or secondary messengers (Gutierrez-Castellanos et al., 2022). At the molecular level, they are also implicated in mediating epigenetic modifications such as chromatin remodeling, altering the accessibility of enhancers and promoters to either facilitate or inhibit transcription (Jaric et al., 2019; Kundakovic and Tickerhoof, 2024).

Progesterone is thought to confine estrogen-induced plasticity to specific time windows by downregulating estrogen receptor expression and suppressing signaling pathways activated by estrogen (Baudry et al., 2013). While this mechanism is more well-established in the endometrium (Selcer and Leavitt, 1988), its role and the mechanisms by which it interacts with or potentially counters estrogenic actions in the brain remain underexplored. Testosterone and its metabolites exert effects in the brain through androgen receptor binding as well as its conversion to estrogen via aromatase (Aspesi et al., 2025). Notably, aromatase expression is enriched in neural circuits relevant to male socio-sexual behaviors (Jennings and de Lecea, 2020; Aspesi et al., 2025).

Receptors for hormones including estrogen, progesterone, and androgens are densely expressed in the brain's highly conserved "social

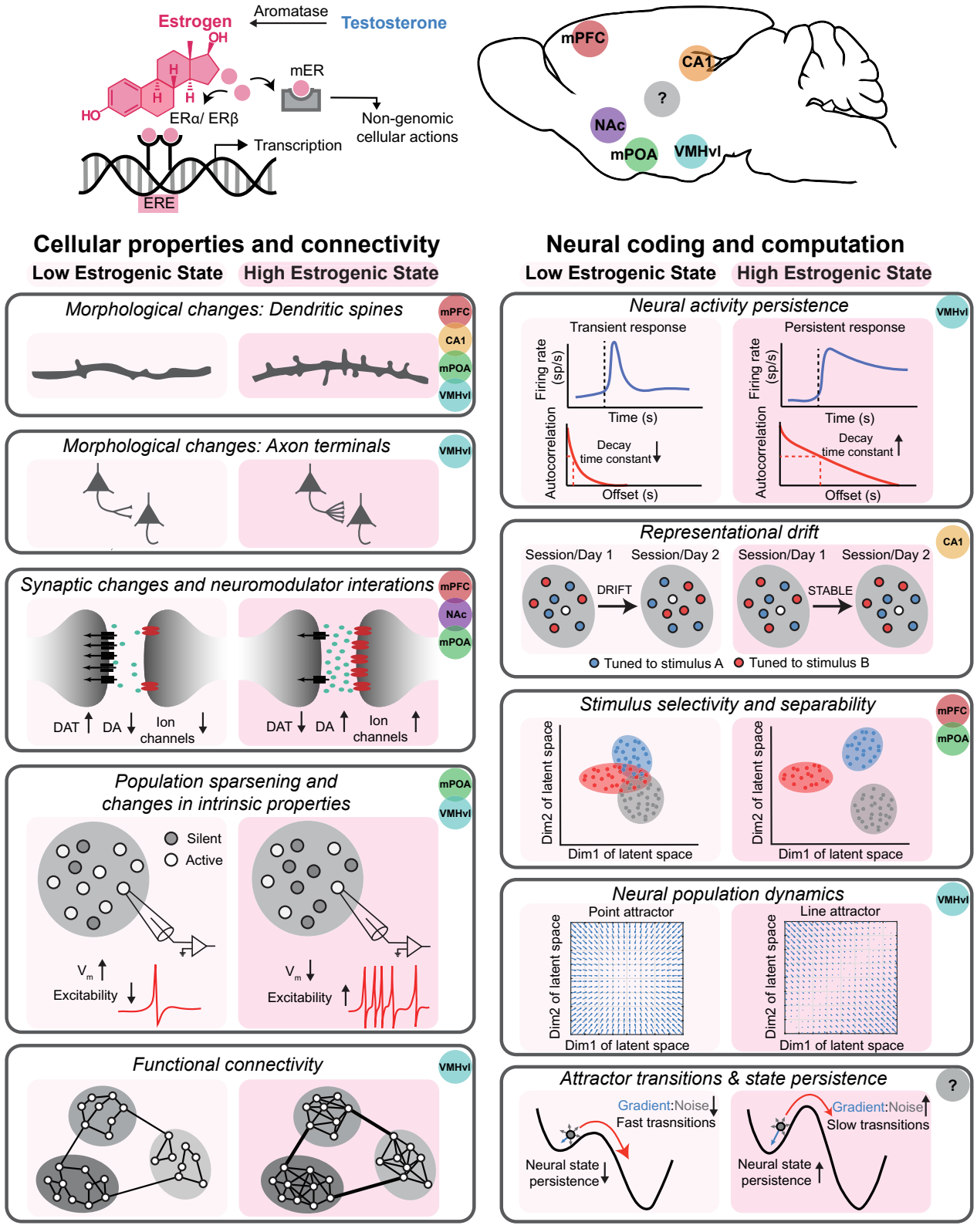


Fig. 1. Sex hormones reshape brain-wide circuits by altering cellular properties, connectivity, neural coding, and computations: (Top panel) Sex hormones bind to specific receptors that regulate gene expression or induce rapid changes through non-genomic intracellular signaling cascades. These receptors are densely expressed in hypothalamic nodes such as the ventrolateral division of the ventromedial hypothalamus (VMHvl) and the medial preoptic area (mPOA), which form the core of the evolutionarily conserved Social Behavior Network. Recent evidence suggests hormone-driven remodeling extends beyond these regions to a broader network involved in social motivation, decision-making, memory, and valence, including the medial prefrontal cortex (mPFC), hippocampus (CA1), and nucleus accumbens (NAc). **(Bottom panel)** Schematic representation of hormone-mediated changes in cellular properties and connectivity (left) alongside alterations in neural coding and computational properties (right) within this extended circuit governing social behavior and cognition.

behavior network" (O'Connell and Hofmann, 2011), a highly connected subcortical network of hypothalamus, amygdala, and midbrain nuclei. At the core of this network are hypothalamic nodes, including the ventrolateral division of the ventromedial hypothalamus (VMHvl), the Medial Preoptic Area (MPOA), and the ventral premammillary nucleus (PMv) (Mei et al., 2023). These regions play a crucial role in driving innate social behaviors by integrating sensory, hormonal, and metabolic cues and providing 'ignition signals' (Stagkourakis, et al., 2023) to downstream brainstem circuits to regulate sexual (Yin et al., 2022; Liu et al., 2024), parental (Stagkourakis, et al., 2023; Ammari et al., 2023), and aggressive (Liu et al., 2022; Stagkourakis et al., 2020) behaviors.

Recent years have also seen a promising uptick in studies investigating the effects of gonadal hormones in circuits for memory, cognition, and decision-making including the hippocampus (Wolcott et al., 2025), nucleus accumbens (NAc) (Golden et al., 2023) and medial prefrontal cortex (mPFC) (Wang et al., 2025). The hippocampus, beyond its role in spatial memory and cognition, is also crucial for encoding social memory and has recently been implicated in detecting social novelty and discriminating between social safety and social threats (Hitti and Siegelbaum, 2014; Kassraian et al., 2024). Recent work shows that fluctuating hormone levels across the estrous cycle shape the dendritic spine dynamics and the stability of place cell tuning in the hippocampus (Wolcott et al., 2025). The NAc, a critical component of the mesolimbic reward system, mediates motivated behaviors, reinforcement learning, as well as social reward seeking (Isaac and Murugan, 2024), with dopamine release in NAc playing a critical role. Recent work elucidates that increased estrogen in rats downregulates the dopamine transporters resulting in slower reuptake and prolonged dopaminergic signaling in the NAc (Golden et al., 2023). The mPFC is a pivotal hub in mediating decision making, integrating diverse streams of information and exerting top-down control over behaviors through extensive long-range efferent projections (Yizhar and Levy, 2021; Le Merre et al., 2021; Mack et al., 2024). It governs several processes underlying social decision-making like social recognition (Levy et al., 2019; Kingsbury et al., 2020), social exploration (Murugan et al., 2017; Liang et al., 2018) and processing social hierarchies (Wang et al., 2011) and social rewards (Isaac and Murugan, 2024). Despite this well established role in social cognition, the role of hormonal signaling in shaping such a critical node is less known and has likely been underestimated due to lower estrogen receptor expression compared to the hypothalamus and hippocampus, but cortical areas still express ERα and ERβ, and hormonal activation mediates changes in gene expression and in intrinsic neuronal properties (Clemens et al., 2019; Mogi, Takanashi, Nagasawa, & Kikusui, 2015; Wang et al., 2025). Recent work demonstrates that a subset of mPFC neurons that project to anterior hypothalamus and express T-type calcium ion channels can track the estrous state and integrate the hormonal state with external social cues (Wang et al., 2025). Beyond these areas, the lateral septum (LS) and amygdala also integrate inputs from the mPFC and hippocampus and modulate social preference, reward, and valence processing (de León Reyes et al., 2023; Gangopadhyay et al., 2021). While these areas also express receptors for several hormones, the role of sex hormones in shaping these circuits remains to be explored.

In addition to modulation of circuits for behavior and cognitive processing, a handful of studies suggest that sex hormones can also modulate sensory processing, potentially gating downstream neural processing. For example, estrogen has been shown to regulate processing of sensory inputs in the olfactory (Dey et al., 2015), somatosensory (Clemens et al., 2019) and auditory pathways (Banerjee and Liu, 2013; Caras, 2013). Even peripheral sensory nodes can alter their stimulus-evoked responses in a hormone-dependent manner (Dey et al., 2015). Since such changes can impact neural computations in downstream circuits that receive inputs from these nodes, targeted perturbations (Vinograd et al., 2024) of specific nodes will be key to determining whether hormones intrinsically remodel these circuits to drive behavioral changes or whether the observed effects are inherited from hormone-mediated changes upstream. Overall, these findings highlight the potential for sex hormones to shape behavior by influencing sensory perception, action generation, and higher-order cognitive processes.

4. Overcoming technical challenges to a systems-level approach to neuroendocrinology

Until recently, our understanding of how changes in hormones regulate social cognition and emotional states has been primarily limited to molecular and cellular mechanisms. There existed a significant gap in the ability to connect molecular processes, gene expression, and cellular changes with population-level dynamics, network-level emergent properties, brain-wide neural activity, and behavior. However, recent technological advances in the ability to 1) record large populations of neurons and 2) track their activity across time have finally made it possible to link hormonal action to network and circuit computation.

The ability to perform large-scale neural recordings with high spatiotemporal resolution in freely moving animals is now relatively routine. Miniaturized, high-density probes like Neuropixels (Jun et al., 2017) and head-mounted "miniscopes" (Aharoni et al., 2019; Zong et al., 2022) now enable simultaneous recording of hundreds or thousands of neurons within a region or across regions. These tools offer valuable insights into population-level activity structure and patterns. In addition, to discern region-specific versus general principles of hormonal action across the brain, the ability to record from multiple regions simultaneously during freely moving behaviors in different hormonal states will be the next frontier. Advances in multi-probe chronic Neuropixels implants (Melin, et al., 2023) and multi-fiber photometry (Iravedra-Garcia, et al., 2024; Dai et al., 2024) now enable this capability.

Tracking neural activity across long periods of time has also been a technical challenge. Steroid hormones typically act on a relatively slow (genomic) timescale, with endogenous hormonal state changes occurring over hours, days, or even longer making it difficult to connect the changes in hormonal modulation and neural activity. At the cellular level, recent developments in analysis pipelines for tracking neuron identity over time (van Beest et al., 2024) now make this tractable. In human imaging, this has been solved through methodological innovations, including "dense sampling", where repeated imaging of the same individual allows researchers to track network reorganization across the timescale of hormonal state change (Pritschet et al., 2024; Pritschet et al., 2020; Grotzinger et al., 2024).

5. Computational and algorithmic frameworks for understanding how hormones generate neural and behavioral persistence

Recent technological advances in spatial and temporal sampling have finally made it possible to discover generalizable principles of hormonal function. Here, we will highlight recent work, from across disparate hormone sensitive circuits for behavior and cognition that track neural changes in neural coding that enable persistence.

5.1. *Increased stimulus sensitivity and separability*

Hormone-dependent enhancements in stimulus sensitivity and separability have been shown to generate highly specific responses to behaviorally relevant stimuli. Sensitivity is a reflection of the tuning of a single neuron, while separability is the uniqueness of this response relative to the population. These generalizable properties have been demonstrated in circuits for behavior and cognition (McHenry et al., 2017; Ammari et al., 2023). In one study (Ammari et al., 2023), pregnant mothers with elevated hormone levels, compared to virgins, exhibit improved separability of pup representations in the MPOA relative to other stimuli such as intruders, food, and objects in a dimensionality-reduced latent space of population neural activity. Estrogen was found

to silence neurons and sparsen population activity while paradoxically increasing excitability. This potentially explains the stronger, more selective, and separable responses. This separability inversely correlates with the fraction of activated neurons and baseline spontaneous activity, suggesting estrogen-mediated sparsening improves signal-to-noise ratio and encoding efficiency.

Hormonal priming using estrogen also enhances sensitivity and gain to social rewards and appetitive cues VTA-projecting MPOA neuron population expressing neurotensin (McHenry et al., 2017). Similarly, the VMHvl shows increased excitability and heightened male-evoked responses during high-estrogenic days in naturally cycling mice (Yin et al., 2022). Increased stimulus sensitivity and separability are also observed under high estrogenic states in a subset of mPFC neurons expressing T-type calcium channels which are under estrogenic regulation (Wang et al., 2025). Population-level analyses reveal enhanced discriminability of male versus female interaction trials in estrus compared to diestrus females (Wang et al., 2025).

5.2. *Stable neural coding and reduced representational drift*

Changes in consistency or stability of neural coding and reduced remapping of tuning over time likely contributes to behavioral persistence. Representational drift, the phenomenon of large-scale changes in neural tuning, is thought to arise from the turnover of spines and synaptic proteins that determine synaptic strength and signaling (Mau et al., 2020). While representational drift supports continual learning and memory updating, it can challenge the maintenance of stable behavior (Driscoll et al., 2022).

In the hippocampus, place cells that encode spatial locations exhibit greater tuning stability when animals are reintroduced to the same environment during the high-estrogen proestrus stage compared to low-estrogen stages (Wolcott et al., 2025). This heightened stability in familiar environments coincides with increased spine density, enhanced dendritic excitation, and stronger somato-dendritic coupling in hippocampal neurons during the high-estrogenic stage (Wolcott et al., 2025).

5.3. *Attractor dynamics*

Persistence in neural dynamics can refer to the ability of a neural population to maintain a specific pattern of activity over time, even in the face of external perturbations. This property is often linked to the concept of attractors, where the neural system is drawn toward a stable state or pattern that can persist over extended periods (Ebitz and Hayden, 2021). Ensemble activity patterns characterized by long-lived neural patterns with right-skewed dwell time distributions align with the dynamics expected from a series of metastable attractors (Mazzucato, 2022; Recanatesi et al., 2022). Modeling the neural landscape as a multi-stable attractor system has provided the computational framework for understanding categorical decision-making (Wang, 2002), how decision stability is maintained in the presence of distractors (Finkelstein et al., 2021) and how conflict between competing needs are resolved to support the persistence of a behavioral state (Richman et al., 2023).

Recent data has pointed directly to the role of sex hormones in enabling this persistence. Dynamical systems analysis of *Esr1*-expressing populations in the female VMHvl uncovered signatures of a line attractor in neural state space during sexually receptive behaviors, specifically on high-estrogen days (Liu et al., 2024). This line attractor, characterized by a dominant dimension of neural activity that decays with a large time constant, is hypothesized to stably maintain a persistently receptive state during interruptions in male copulatory behavior. Moreover, the ramping dynamics of this attractor are thought to encode the escalating sexual arousal of the female.

6. Hormone-mediated changes in functional connectivity and communication

As a mechanistic link between hormone-mediated cellular change and population coding, mounting evidence across species has suggested that hormones can alter fundamental properties of circuit communication, including shifts in recurrent connectivity (Vinograd et al., 2024) or slow neuromodulation (Mountoufaris et al., 2024).

Estrogen is known to broadly facilitate synaptic potentiation across various brain regions (Gutierrez-Castellanos et al., 2022). In a direct measure of changes in long-range circuit communication, estrogen has been shown to increase the density of female VMHvl axon terminals in the anteroventral periventricular (AVPV) nucleus, a hypothalamic node involved in sexual receptivity (Inoue et al., 2019). This structural plasticity improves synaptic transmission efficiency and functional connectivity between these regions.

Morphological changes associated with high-hormone levels, including increased dendritic spine density, have been documented in the hippocampus (Wolcott et al., 2025), mPFC (Ye et al., 2019), VMHvl (Stagkourakis et al., 2020; Calizo and Flanagan-Cato, 2000), and MPOA (Ammari et al., 2023). These changes provide the structural basis for enhanced neural communication, potentially leading to more efficient and flexible network activity. Hormone-mediated increases in functional connectivity have also been observed in human neuroimaging studies (Pritschet et al., 2024; Pritschet et al., 2020; Grotzinger et al., 2024). Further research combining large-scale neural recordings with techniques to manipulate and monitor connectivity is essential to elucidate how these structural and functional adaptations causally contribute to hormonally driven stable behaviors.

Finally, gonadal hormones can exert profound effects on neural activity and communication by interacting with other neuromodulators, including dopamine (Golden et al., 2023; Yoest, Quigley, & Becker, 2018), norepinephrine (Bangasser, Wiersielis, & Khantsis, 2016), oxytocin (Quintana et al., 2024), and vasopressin (Aspesi and Choleris, 2022). For example, estrogen enhances sensitivity to reward values in female rats by driving better reward prediction errors and reinforcement learning during high-estrogenic states (Golden et al., 2023). This effect is mediated by estrogen-induced reductions in dopamine transporter levels in the NAc, which decreases dopamine reuptake and leads to elevated dopamine levels. Estrogen priming is also believed to enhance sensitivity to oxytocin and is associated with increased parental behaviors (Quintana et al., 2024). In the mouse mPFC, oxytocin receptor-expressing inhibitory interneurons have been implicated in mediating critical inhibitory transmission that likely improves stimulus selectivity and signal-to-noise ratio, contributing to better behavioral and perceptual outcomes during high-estrogenic states (Nakajima, Görlich, & Heintz, 2014; Wang et al., 2025). Slow neuromodulation through oxytocin and vasopressin-mediated neuropeptide signaling is also essential for maintaining specific neural dynamics. For example, these neuromodulators implement line attractor dynamics in VMHvl (Mountoufaris et al., 2024), and these attractor dynamics further play a causal role in maintaining persistent internal states (Vinograd et al., 2024). Understanding these generalizable functions, including changes to network computation and communication, will continue to provide critical insights into the integrative roles of gonadal hormones and neuromodulators in shaping neural and behavioral dynamics.

7. Open questions and future directions

While recent work has made substantial progress in tying hormonal modulation to computational functions, many open questions remain. Future work must disentangle the causal effects of hormones from purely correlative associations. This will include mapping the diversity of gene expression to changes to neural functioning. In addition, a major open question is understanding how multiple hormones with different time-varying processes can operate synergistically to change neural

computation and behavior. To answer this, region-specific manipulation of specific hormone receptor expression provides a powerful approach to investigate the causal role of hormone-driven remodeling in specific brain regions in mediating changes in targeted behaviors. For example, genetic knockdown of estrogen receptors specifically in VMH leads to deficits in female social behaviors (Musatov et al., 2006), as well as dysregulated metabolism and body weight (Musatov et al., 2007). Similarly, estrogen receptor knockdown in midbrain VTA suppresses reward sensitivity and reinforcement learning (Golden et al., 2023). Conversely, overexpressing estrogen receptors in NAc mimics the resilience-associated transcriptional signatures and enhances stress resilience (Lorsch et al., 2018). Future work combining site-specific genetic manipulations of targeted hormone receptors with large-scale neural recordings will be essential for determining whether hormone-driven local transcriptional changes also causally influence neural dynamics and computations. In sum, a systems-level approach to neuroendocrinology will be critical to understanding how sex hormones shape and reshape neural dynamics to prioritize and sustain specific behaviors.

CRediT authorship contribution statement

Meenakshi M. Asokan: Writing – review & editing, Writing – original draft, Conceptualization. **Annegret L. Falkner:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.nlm.2025.108064.

Data availability

No data was used for the research described in the article.

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